

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460616.777 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.011
Transit depth (Rp/Rs)^2	2.21 +/- 0.33 [%]
Orbital Inclination (inc)	84.18 +/- 1.72
Airmass coefficient 1 (a1)	1.1577 +/- 0.0068
Airmass coefficient 2 (a2)	-0.133 +/- 0.05
Scatter in the residuals of the lightcurve fit is	0.58 %
Best Comparison Star	None
Optimal Aperture	4.37
Optimal Annulus	10.44
Transit Duration (day)	0.1003 +/- 0.0052

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.149 $\pm$ 0.011 vs 0.1489 (deviation 0.01 σ)
Duration fit vs published	0.1003 d vs 0.1304 d (deviation 5.79 σ)
Mid-time O-C	9.4 min
Depth SNR	13.5
Residual scatter	0.58 %

Light curve

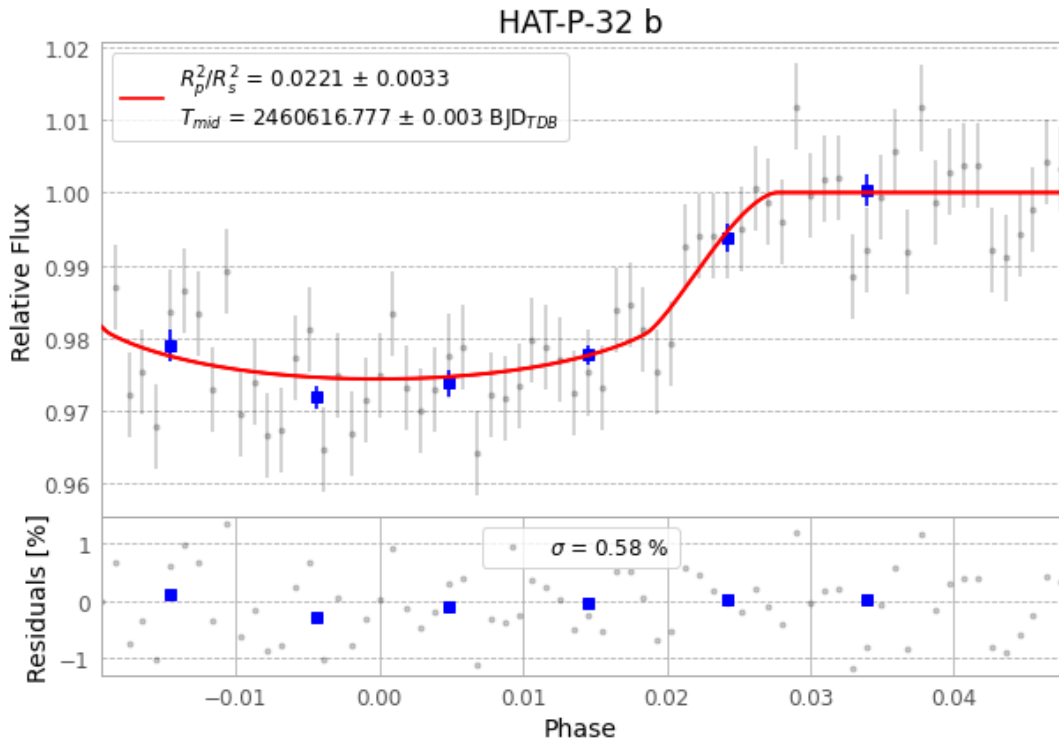


Figure 1 — FinalLightCurve_HAT-P-32 b_2024-11-01.png (SHA-256 b3c445d5a44e4e47...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [392, 239], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dcd39d6c7290a384...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 378b18d

Data provenance

70 FITS frames (46 MB), HATP-32241102053016.FITS → HATP-32241102085715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-241102.log (c5fe885323aa...), FinalLightCurve_HAT-P-32 b_2024-11-01.png (b3c445d5a44e...), FinalLightCurve_HAT-P-32 b_2024-11-01.csv (55a2d8919929...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 21:08Z.